

Task 2 — Project Report

Feature Engineering & Model Comparison for House Price Prediction

Dataset:

California Housing Dataset (Scikit-Learn)

Models:

Linear Regression, Ridge Regression, Decision Tree Regressor

Language:

Python 3.14

Libraries:

Pandas, NumPy, Scikit-Learn, Matplotlib

1. Objective

The objective of this project is to perform feature preprocessing and compare multiple machine learning models for predicting house prices using the California Housing Dataset.

2. Dataset Overview

The California Housing dataset contains 20,640 housing records with 8 input features.

Feature	Description
MedInc	Median Income
HouseAge	House Age
AveRooms	Average Rooms
AveBedrms	Average Bedrooms
Population	Population

Feature	Description
AveOccup	Average Occupancy
Latitude	Latitude
Longitude	Longitude
Target Variable:	
MedHouseVal (House Price)	

3. Data Preprocessing

The following preprocessing steps were performed:

- Data loading
- Data exploration
- Feature scaling using StandardScaler
- Train-Test split (80:20)
- Feature preparation

Training samples: 16,512

Testing samples: 4,128

4. Models Implemented

Linear Regression

A basic regression model used as a baseline.

Ridge Regression

An improved regression model that uses regularization to reduce overfitting.

Decision Tree Regressor

A tree-based model that predicts values using decision rules.

5. Model Evaluation

The models were evaluated using:

- RMSE (Root Mean Squared Error)
- R^2 Score

Model	RMSE	R^2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree	0.724234	0.599732

6. Feature Importance

Feature importance analysis was performed using the Decision Tree Regressor.

The analysis indicated that some features contributed more significantly to house price prediction than others.

7. Results

- Decision Tree achieved the lowest RMSE.
- Decision Tree achieved the highest R^2 score.
- Ridge Regression performed similarly to Linear Regression.
- Decision Tree produced the best overall performance.

8. Conclusion

This project successfully implemented multiple machine learning models for house price prediction.

Among all models, the Decision Tree Regressor achieved the best performance with:

- RMSE = 0.724234
- R^2 Score = 0.599732

Therefore, Decision Tree Regressor was selected as the final model.